

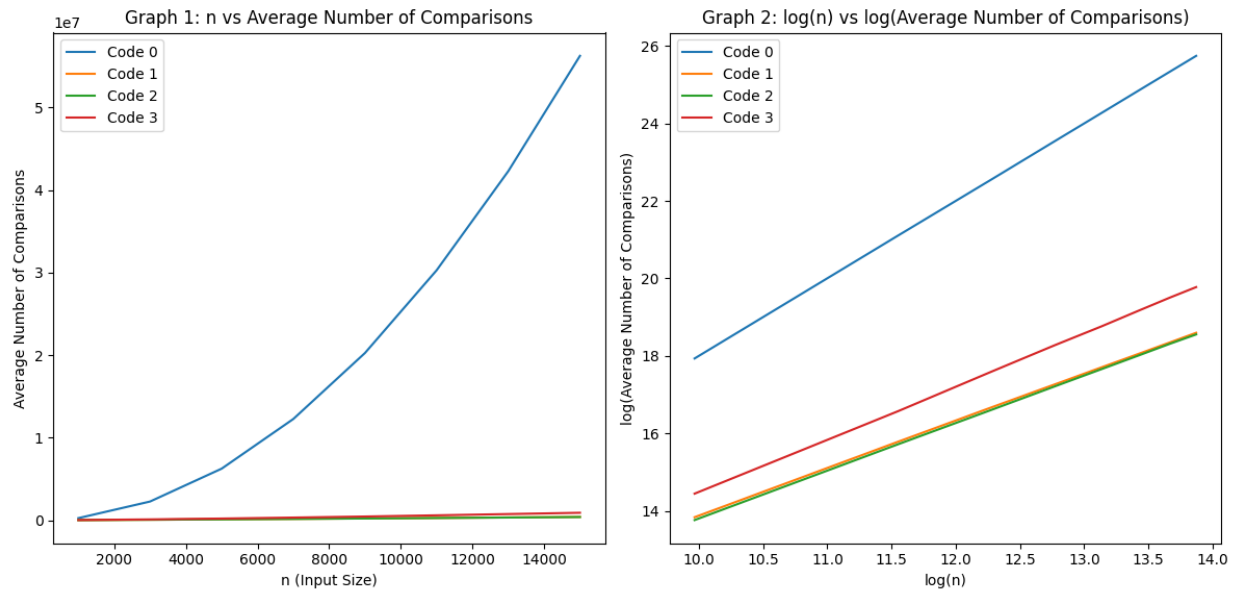
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Program 1

Part II: Analysis



According to the graphs, Code 2 runs the fastest as it has the least amount swaps, regardless of n , input size. Code 2 is also the most efficient asymptotically as it can handle both smaller size arrays and large sizes (up to 15,000 numbers in the array) based off the graphs. Early on, Code 2 reduces the amount of swap operations by avoiding the small gaps early on, minimizing the redundant movements in the sort. Because of this, it scales and performs better than Code 1, which is less efficient with minimal differences.